

AI Intern Assignment

Scope

A student wants to do a PhD. We want to send cold emails on their behalf to the right professors — emails that earn replies and turn into conversations that lead to offers.

Input the system gets:

- The student's profile (CV, transcript, papers if any)
- A description of their research interests in their own words

Output the system produces, end-to-end:

- The list of professors / labs to target (the system picks them — the student doesn't supply names)
- The first outbound email sent to each one

Constraint: fully automated. No human in the loop between the student handing over their profile and the emails going out. Not the student, not us, not an editor. If your design needs a human anywhere in that path, you've changed the assignment — call that out and defend it.

Out of scope: reply handling, the application itself, anything after the first email.

Goal

Maximize the rate at which these emails lead to a real conversation with a professor the student would want to work with.

How you define "real conversation" and "would want to work with" is part of the assignment.

What we want from you

A short write-up of how you'd approach this. We care about:

- **The system you'd build.** What it does, end-to-end. Surprise us — the obvious decomposition isn't necessarily the right one.
- **The trade-offs you're choosing.** Pick a side. Don't list considerations without ranking them.
- **How it gets better over time.** A v1 that doesn't learn from its own outcomes is a dead system. What's the feedback loop? What signal do you collect, from whom, and how does it change what the system does next month?
- **What you'd ship first.** The smallest version you'd put in front of ten real students, and what you'd measure to know it worked.

Deliverable

Two things:

1. **An architecture write-up.** How the system works end-to-end, why this design over the alternatives you considered, where it can fail, how it improves over time. Prose, sketches, examples — whatever makes your case. No length target.
2. **A simple MVP.** A runnable implementation of the core flow. Doesn't need to be production-quality or cover every edge case — pick the smallest slice that proves the idea works on at least one real student profile end-to-end, from input to drafted email(s) for self-chosen targets. Ship the code with a short README on how to run it and what you cut.